$$y = a + bx$$

where a and b are constants. Any value of a and b would give a straight line, and once these values are obtained an estimate of y can be obtained by substituting the observed values of y. In order that the equation $y = a + b\,x$ gives a good representation of the linear relationship between x and y, it is desirable that the estimated values of y_i, say $\hat{y}_i$ on the whole close enough to the observed values y_i, $i = 1, 2, \ldots, n$. According to the principle of least squares, the best fitting equation is obtained by minimizing the sum of squares of differences $\sum_{i=1}^{n}\left(y_i - \hat{y}_i\right)^2$

That is, $\sum\left(y_i - \hat{y}_i\right)^2 = \sum_{i=1}^{n}\left(y_i - a - bx_i\right)^2$ is minimum. This leads us to two normal equations.

$$\sum_{i=1}^{n} y_i = na + b\sum_{i=1}^{n} x_i \tag{7.1}$$

$$\sum_{i=1}^{n} x_i y_i = a\sum_{i=1}^{n} x + b\sum_{i=1}^{n} x_i^2 \tag{7.2}$$

Solving these two equations we get the vales for a and b and the fit of the trend equation (line of best):

$$y = a + bx \tag{7.3}$$

Substituting the observed values x_i in (7.3) we get the trend values y_i, $i = 1, 2, \ldots, n$.

Note: The time unit is usually of uniform duration and occurs in consecutive numbers. Thus, when the middle period is taken as the point of origin, it reduces the sum of the time variable x to zero $\left(\sum_{i=1}^{n} x_i = 0\right)$ and hence we get

$a = \dfrac{\sum_{i=1}^{n} y_i}{n}$ and $b = \dfrac{\sum_{i=1}^{n} x_i y_i}{\sum_{i=1}^{n} x_i^2}$ by simplifying (7.1) and (7.2)

The number of time units may be even or odd, depending upon this, we follow the method of calculating trend values using least square method.

Merits

- The method of least squares completely eliminates personal bias.
- Trend values for all the given time periods can be obtained
- This method enables us to forecast future values.

Demerits

- The calculations for this method are difficult compared to the other methods.
- Addition of new observations requires recalculations.
- It ignores cyclical, seasonal and irregular fluctuations.
- The trend can be estimated only for immediate future and not for distant future.

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